October 24, 2026

The Editor-in-Chief
Measurement

Dear Editor,

We are pleased to submit the manuscript entitled "Assessing the Similarity of Acoustic Signals Using a Vector Representation of Sound Features – From Acoustic Features to Perceptual Difficulty: How to Compare Two Sounds?" for consideration as a Research Paper in *Measurement*.

This is a resubmission of manuscript MEAS-D-26-21300, which was returned to us before review. As requested, we now provide an editable (LaTeX) version of the manuscript prepared in accordance with the Guide for Authors, and this revised cover letter is addressed to *Measurement*. We apologise for the error in the previous cover letter.

The work addresses a measurement problem: how to quantify, objectively and reproducibly, the similarity of two acoustic stimuli. Auditory perception and cognitive tests that present several sounds simultaneously assume that task difficulty reflects the ability of the listener. If two stimuli are acoustically very close, however, the test partly measures the properties of the sound material instead. A quantitative measure of stimulus similarity is therefore needed to control this source of error.

We describe each of 24 animal sounds by a 16-dimensional vector of measured acoustic features (spectral centroid, fundamental frequency, peak-to-average ratio and 13 mel-frequency cepstral coefficients). We apply min–max normalisation so that parameters expressed in different units contribute comparably, and we use the Euclidean distance between the normalised vectors as the similarity measure. Correlation analysis confirms that the selected descriptors carry non-redundant information, and principal component analysis is used to visualise the feature space. The measure is then compared with measured listener performance from the Memory Load Test (MLT) study: the stimulus pair with the smallest distance ($d = 0.736$) was identified correctly in only 20.5% of trials, whereas the more distant pairs reached 44.4–92.9%. The relationship was not monotonic, which we report and discuss openly as a limitation requiring further validation.

We believe the manuscript is relevant to the readers of *Measurement* because it proposes a transparent and reproducible procedure for the quantitative characterisation of acoustic signals, and it relates this physical measure to the measured performance of listeners. In practice, the procedure allows stimulus sets for audiological and cognitive testing to be screened for overly similar pairs before use, which improves the standardisation of the test material. The feature data and the analysis script have been deposited in Zenodo to support reproducibility.

We confirm that this work has not been published previously, is not under consideration for publication elsewhere, and that its publication has been approved by all authors.

Yours sincerely,

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